Tutorial 2

Prove the following statement.

Theorem (Hessian test for convexity). Let $C \subseteq \mathbb{R}^d$ be open and convex, and let $f : C \to \mathbb{R}$ be twice continuously differentiable. Then

$$f \text{ is convex on } C \quad \Longleftrightarrow \quad \nabla^2 f(x) \succeq 0 \qquad \text{for every } x \in C.$$

1. **Conceptual checks**

   Determine whether each statement is *True* or *False*. Give a short justification; for a false statement, give a counterexample.

   (a) If $f : \mathbb{R}^d \to \mathbb{R}$ is strictly convex and differentiable and $\nabla f(x^*) = 0$, then x^* is the unique global minimizer.

   (b) If $f : \mathbb{R} \to \mathbb{R}$ is twice continuously differentiable, $f'(x^*) = 0$, and $f''(x^*) \geq 0$, then x^* must be a local minimizer.

   (c) For a μ-strongly convex and L-smooth function, gradient descent with step size $1/L$ has rate factor $1 - \mu/L = 1 - 1/\kappa$, where $\kappa = L/\mu$. Thus a larger condition number gives a slower worst-case rate.

2. **Logistic regression: strict versus strong convexity**

   Let

   $$a_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad y_1 = 1, \qquad a_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad y_2 = -1,$$

   and

   $$\ell(x) = \frac{1}{2} \sum_{i=1}^{2} \log\bigl(1 + e^{-y_i a_i^T x}\bigr), \qquad x \in \mathbb{R}^2.$$

   Set $m_i(x) = y_i a_i^T x$ and

   $$w_i(x) = \frac{e^{m_i(x)}}{(1 + e^{m_i(x)})^2}.$$

   You may use

   $$\nabla^2 \ell(x) = \frac{1}{2} \sum_{i=1}^{2} w_i(x) a_i a_i^T.$$

   (a) Show that $\nabla^2 \ell(x) \succ 0$ for every finite x, and deduce that ℓ is strictly convex.

   (b) For $x_t = t(1, -2)^T$, show that ℓ is not strongly convex and that $\inf \ell = 0$ is not attained. Explain why this does not contradict strict convexity.

3. **Conditioning and gradient descent**

For $\kappa \geq 1$, consider

$$f_\kappa(x) = \frac{1}{2}\big(x_1^2 + \kappa x_2^2\big), \qquad x = (x_1, x_2)^T \in \mathbb{R}^2.$$

(a) Determine the sharp strong-convexity constant μ, the sharp smoothness constant L, and the condition number.

(b) Run gradient descent with step size $\alpha = 1/L$. Derive the coordinate-wise update and hence an explicit formula for $x^{(k)}$ in terms of $x^{(0)}$.

(c) Explain, from the coordinate formula, why a large condition number slows gradient descent even though the high-curvature coordinate is reduced very quickly. Relate your explanation to the rate factor $1 - \mu/L$.